

CHAPTER 12

Discrimination and Male–Female Earnings Differentials

Answers to End-of-Chapter Questions

1. (LO1) This statement is false. There is at least one model of labour-market discrimination that is totally neoclassical and includes a discriminatory wage equilibrium. The crowding hypothesis works by shifting the supply of female labour down and to the right for many industries and/or occupations that are female-dominated, and simultaneously shifting the supply of female labour up and to the left for industries and/or occupations that are male-dominated. In the crowded sectors where many women are pushed, one moves down and along the demand curve as this expansion in supply occurs, which means that the female workers are paid the marginal revenue product of their labour. Their wage is less than the wages of equally qualified males, because the male-dominated occupations and industries are subjected to the flipside of the supply push. In those labour markets, the crowding results in higher wages than would prevail if there were no barriers to entry into the male-dominated industries and/or occupations.

2. (LO1) This statement is also false. There are a number of economic mechanisms by which discrimination could persist in the long run using non-competitive theories. The first such mechanism is the one associated with labour force adjustment costs. According to the Arrow proposition, firms which have discriminated in the past are reluctant to undo the wage discrimination that occurred because it would involve firing a lot of their male workers and/or cutting their pay, both of which can wreak havoc on labour relations. A related story is the efficiency wage hypothesis, according to which firms pay super-competitive wages in order to reduce turnover and elicit more effort on the part of their workers. This phenomenon goes hand-in-hand with unions working to raise wages, and other non-competitive wage-employment outcomes. When super-competitive wages are paid (which include economic rents), there will be workers queuing for these scarce jobs. Under such conditions, discrimination against women and minorities is more likely. According to this proposition, the incumbent employees, who are mostly male, benefit from the discrimination, and place pressure on the employer (and perhaps on female applicants) in the act of hiring them.

The theme that links these three propositions together is that firms are more likely to maximize profits when the morale of the labour force is high, and when turnover is fairly low. In colloquial terms, employers are rarely spoiling for a fight with their incumbent employees. Labour peace and conscientious work may come at a price of discrimination, and the competitive pressure that normally emerges in instances of wage differentials is contravened.

Although the monopsony model is a little hard to apply, it constitutes another case where wage discrimination can persist in the long run.

3. (LO1, 2) This is a somewhat controversial question. For purely neoclassical economists who believe that most individuals make economic choices according to their own free will, the answer is yes. They tend to claim that most of the wage differential between men and women can be attributed to productivity differences. According to this perspective, women tend to have lower marginal productivity levels because men tend to have higher rates of labour-force attachment and somewhat higher levels of training and education. Higher wages for men are justifiable on this basis.

A less orthodox economist would disagree with that perspective. The central thrust of the counterargument is that a worker's marginal productivity level is not always determined by the choices of the worker (such as how much human capital to obtain) and his/her natural endowment (innate characteristics, talent). The human capital decision (which in women's cases is sometimes affected by childbearing) may be a matter of rational, individual choice, or it may also be a result of discriminatory social practices. It is true that on average, women have higher turnover and absenteeism rates than men do, and this is often tied to household responsibilities.

A less-than-orthodox economist would then pose the question of why this practice emerges. Do women have equal opportunities to acquire human capital, the degree of labour-force attachment, or the level of experience that many employers require in order to pay high wages? Some observers would say no. This point is analogous to the "Which came first, the chicken or the egg?" question.

4. (LO6) The laws will not little impact on the wages and employment of female workers. These laws only apply in situations in which it is recognized that the jobs within a particular firm performed by men and women are similar. A sanction can only be implemented following an appeal by a plaintiff. Theoretically, one would expect equal-pay laws (in those few instances in which they apply) to reduce the employment of women. This flows from a simple application of the law of demand. In the text, it is stated that the limited empirical evidence that does exist suggests that these policies are ineffectual in closing the wage gap. What impact would equal employment opportunity laws have on the wages and employment of female workers? Such laws are designed to bring about an expansion of the demand for labour of women. If such laws succeed, the wage and the employment level of women should increase. As the textbook notes, this is more likely to benefit women entering the labour market than incumbent female workers. The empirical evidence is discussed briefly in the paragraph entitled "Affirmative Action." Affirmative action seems to have been successful in improving labour market outcomes for whatever group is targeted, although it may impose costs on other groups of workers.

Both of these policies, by the way, have a very positive effect on the wages and employment levels of lawyers who practise in this area of employment law. There is little question that it consumes the economic resources of both firms and employee groups, and

thus reduces employment in the aggregate.

5. (LO5) For policy purposes, does it matter if the reason for discrimination is prejudice, erroneous information (systemic discrimination), or job security? Yes, it matters. For the case of prejudice, one possible policy response is to attempt to alter preferences and attitudes through public relations campaigns and sensitivity workshops. For the job security factor, equal-opportunity employment policies designed to increase the number of openings for women may be warranted in some cases, as the discrimination is tied to occupational segregation. The proponents of employment equity and comparable worth/pay equity claim that it is an effective instrument to combat erroneous information that gives rise to discrimination. Direct control and regulation are viewed by some as the most effective way to compel employers to hire targeted groups, regardless as to whether employers intentionally discriminate or not. For the erroneous information type of discrimination, sometimes employers engage in systemic discrimination because they believe that female workers will be absent frequently. Policies to facilitate female employment might be the most productive. An illustration would be improving day care services and the flexibility of hours, which would help meet the needs of female employees and the needs of their employers (less absenteeism). This is the type of policy measure that gains the most favour among economists.

Does it matter if the main source of discrimination is employers, co-workers, or customers? Yes, it does, and the latter is the most difficult to combat. If the source of the discrimination lies with the firm's customers, it almost absolves the firm of responsibility. One can initiate employment equity proceedings against a firm whose owners or workers are alleged to engage in discriminatory employment practices, but one cannot take any kind of action against customers who discriminate. Facilitating policies are probably the least controversial method of combating discrimination, but they are ineffective against the source of customer prejudice.

6. (LO1) It is because their family situations are probably very similar. In both cases, there are no childbearing responsibilities at all. One major factor, but certainly not the only factor, behind the male-female wage gap is that women lose a lot of labour market experience relative to men, to the extent that they are the primary caregivers of the children. If we are talking about single individuals as well as childless individuals, both men and women are likely to be highly attached to the labour force, and thus both are likely to have comparable levels of labour market experience. The labour supply behaviour should be very similar.

If this gap is negligible, one should control for human capital factors and other pre-market characteristics before claiming that there is no discrimination based on a comparison of average male and female salaries. For example, if a group of women have higher levels of formal education than a comparable group of men, and the unadjusted pay gap is low, then there might be discrimination against women because they ought to be earning more in the particular situation. One might also want to control for the sectoral

or occupation composition of the comparison groups of men and women as well, assuming that those are pre-market characteristics of the workers. Having said that, a crucial issue always arises: to what extent are the sectoral and the occupational distributions of women workers affected by discrimination?

7. (LO5, 6) See the subsections entitled “Scope of Equal Value Initiatives” , “Design Features of Comparable Worth”, “Pay Equity Example”, and “Comparable Worth and Pay Equity”, as well as Exhibits 12.12 and 12.13. One important factor is that comparisons are usually made within the same employer. It is for this reason that, in practice, pay equity initiatives have been carried out only for very large employers like the federal government and Bell Canada. It is also a complex and costly system to administer. One has to define what constitutes a male-dominated occupation and a female-dominated one, the job evaluation procedure for calculating the values of jobs, the procedure for linking the job evaluation point scores to the rates of pay of jobs (this element is particularly problematic), and the procedure for adjusting the pay in undervalued jobs. This procedure can be very contentious when applied. When the pay equity award is made, it must be financed by the employer, who (depending on the wage elasticity of labour demand) is likely to reduce his/her quantity of labour demanded for the lower-valued occupation. The morale of those workers who are not covered by the package can be undermined.
8. (LO3) This question refers to what the textbook calls the “raw” wage differential. It is decreasing over time, and this is what we would expect to occur. The unadjusted male-female pay gap reflects the influences of both the differences in the characteristics of the two groups of workers (the productivity-related characteristics) and the differences in the returns to those characteristics in terms of earnings growth given the same characteristics (including a reduction in discriminatory pay practices). As women have become more skilled and more educated relative to men, the gap has shrunk. As their labour force participation rate has increased, their experience levels have increased relative to that of men, which has also contributed to closing the gap.
9. (LO5) This statement is true. It surmises the underlying philosophy of the comparable worth approach. The whole concept of comparable worth is antithetical to a market (supply and demand) framework. The contrast between these two underlying paradigms and philosophies is nicely explained in the textbook. Within the comparable worth framework, there is a role for market forces, to the extent that compensating differentials are allowed for adverse working conditions, and differentials are allowed for educational attainment and skill acquisition. Overall, however, the equal value concept rejects the notion that market forces should be the primary determinant of the value, and hence the pay levels, of jobs. According to this perspective, market forces are the cause of occupational segregation, labour market crowding, and the systematic undervaluation of female-dominated jobs. The primacy of market forces is rejected in that the value, and hence the remuneration, of a job is not deemed to be low simply because there is an abundance of labour willing to do the work, or because there is little demand for that type

of labour. These market forces are not judged to produce socially acceptable outcomes. This is in contrast to the economic emphasis placed on the forces of supply and demand to determine the value and the remuneration of a job. Note also the distinction between the administrative concept of value (used in comparable worth) and the economic concept of value (used in the free market).

10. (LO5) The material which follows discusses the likely implication of the design and implementation features of comparable worth, without going into the pros and cons of each point.
- a) *Defining gender dominance as 60% as opposed to 70% of either sex:* This would increase the scope of the policy measure, as more industries would be considered female-dominated industries and thus a candidate for underpaid status and a wage adjustment. More industries would also be considered male-dominated, and hence the average pay level of these highly paid industries, to which the female-dominated industries are to be compared for a possible raise, will fall. This change does not seem logical, because if the threshold were changed to 50%, many industries would be neither male-dominated nor female-dominated.
 - b) *Constraining the male and female pay lines to have the same slope as opposed to different slopes:* This implies that the vertical distance between the two lines is constant. Regardless of the number of job evaluation points, all female-dominated industries would receive the same raise in absolute terms. In percentage terms, female-dominated professions, such as nursing, would get a much smaller comparable-worth raise.
 - c) *Allowing projection of the male payline outside of the sample range of the data in cases where the female jobs otherwise have no male comparator groups:* The implication is that these lower-end female-dominated jobs, with low-level job evaluation points, would receive the same raise, in absolute terms, as all of the other female-dominated jobs would receive. Those latter female-dominated jobs would have received raises based on higher-paying male occupations. Without such an extrapolation, there would be no basis to award a salary adjustment to those female-dominated jobs without a male comparator group according to this comparable worth scheme.
 - d) *Following a point-to-line versus a line-to-line versus a point-to-point wage adjustment procedure:* This question goes into a great level of minute, technical detail. See page 372 for a description of the various possibilities.
11. (LO4, 5, 6) One might expect discrimination (in one form or another) to be more prevalent in the private sector and the non-union sector than in the public or the union sector. The textbook does not give reasons in the case of unions, but *within* the union sector, there is a much wider scope for administered wage setting mechanisms as opposed

to purely market determined wage outcomes. (This does not exclude the possibility that unions can contribute to discrimination *between* union and non-union workers by excluding the latter from choice jobs.) This contrast between administered wages and market-based wages is important. In two places, the textbook mentions that in Australia, where the wages for a majority of the labour force are determined by wage tribunals, gender differentials are much narrower than in Canada, where the wages of most of the labour force are determined by market forces.

The text stated that almost all of the well-known cases of pay equity and comparable worth interventions have occurred in the public sector. All of the various remedies, listed in question 12, are probably easier to implement in the public sector for several reasons. First of all, public sector employers tend to be larger in size, which facilitates the implementation of most of the possible remedies. Second, it is the government which drafts the anti-discriminatory measures, and it has the authority to enforce them and sanction unlawful behaviour. It should thus set an example for private sector employers to follow, and indeed this has occurred. The summary of the history of these various measures indicates that they typically applied only to the public sector during earlier phases.

12. (LO6) The textbook talks about 6 different types of measures. They are presented below with a brief commentary.
- i. *Conventional pay equity*: The textbook mentions a few times that the scope for this measure is small, as it is limited to a comparison of jobs within a firm, and the jobs have to be very similar. Furthermore, most wage discrimination does not take this form. Wage differentials across very similar jobs within the same firm are small.
 - ii. *Equal employment opportunity*: This is not a results-based policy measure, as hiring decisions and wage decisions are not dictated to the employer. It typically seeks to alter hiring and recruitment practices which are viewed to be gender biased. It is costly to enforce, but properly implemented, it should benefit certain groups of women, improving their job opportunities. Although it has the effect of increasing the demand for female labour, it is less helpful for incumbent female employees than it is to women entering the labour force.
 - iii. *Comparable worth/equal value*: Comparable worth as it is currently practised does not involve comparisons across different firms and industries, so the scope is still pretty limited. It cannot reduce the portion of the overall earnings gap that reflects the segregation of females into low-wage establishments or industries or occupations. In most Canadian provinces, it only applies to the public sector. In practice, wage adjustments have not been large. Another factor which matters is the distinction between a complaint-based initiation and a proactive initiation. The latter gives rise to a much wider scope.

- iv. *Employment equity/affirmative action policies:* These are results-oriented, in that employers are strongly encouraged to hire certain numbers of disadvantaged groups. The proponents consider this measure a very explicit way to favour targeted groups who have suffered from systemic discrimination in the past.
 - v. *Policies to facilitate female employment:* These include improving the provision of more flexible work arrangements, part-time jobs, child-care leaves and day care arrangements. This approach assumes that women are assigned a very disproportionate share of family and childrearing responsibilities, and it is not results-oriented at all. As the textbook mentions, to a large extent, they are emerging as endogenous responses to the increased labour market role of women. To the extent that their emergence is hampered by discrimination and other impediments, there is scope for government activity. Some argue in favour of greatly expanded subsidies for day care.
 - vi. *Alter the preferences and attitudes:* In the very long run, governments are capable of shaping certain societal values to a certain extent, although it is difficult to attribute such a change to specific interventions. The success or failure of such policies can only be measured over decades.
13. (LO6) This point is explained explicitly in the subsections of the textbook called “Design Features of Comparable Worth” and “Pay Equity Example” According to the policy-capturing approach, the pricing mechanism for male-dominated occupations is applied to the female-dominated occupations. This procedure is seen to respect, for the most part, the market forces that operate in male dominated occupations. The pricing mechanism for the job evaluation points — the educational requirements, the working conditions and other non-pecuniary factors — is respected and extended to female-dominated jobs. It is the pricing mechanism of the female-dominated jobs that is discarded. This is viewed as a rational outcome, because the market forces are respected, and perhaps even strengthened if the female dominated occupations are subject to non-competitive forces. It is also viewed as an equitable outcome, as the same pricing rules apply to both sets of occupations. This analysis, however, is based on a premise that the job evaluation procedure is very sound and thorough. It is reasonable to expect employers to always resist pay equity programs, as they will always raise labour costs considerably. This is because they require that parity be achieved by raising the wages of females rather than lowering the wages of males. See the response to question 9 for a similar discussion.
14. (LO5) This question goes into great technical detail on the topic of comparable worth. Rather than give a substantive response to each of these points, the solution provided here is limited to one of the lessons this question is designed to illustrate. The main point is that while the objective of comparable worth is clear (to combat the most pervasive form of discrimination of segregating workers into male and female dominated occupations), the mechanics of implementing this procedure are complicated and not immediately

apparent to even the proponents. Only those with substantial exposure to the task can comprehend it. In colloquial terms, the devil is in the details. There are two primary steps: the determination of the number of job evaluation points [the initial job assessment], and the movement from the female-dominated occupation line (or point) to the male-dominated occupation line (or point) [the comparison that is used for the adjustment award].

15. (LO5) Such a policy measure might have the impact of raising the cost of hiring both men and women who are parents, or have child-care responsibilities, relative to men or women who are either childless or are discharged from child-care responsibilities. Firms might be tempted to pass on a fraction of the cost that they incur by providing these services by either cutting the pay of the users, or else reducing the number of such employees that they employ. According to an economic perspective, if the firm operates in a competitive output market, one should expect some type of economic impact. The family-friendly practices will not come without any cost.
16. (LO3) The empirical evidences on male-female pay differentials are discussed in the subsection titled “Additional Evidence on Male–Female Pay Differentials” in the text. The declining gap likely reflects a variety of factors: more accumulation of experience and continuity of experience on the part of women; increases in the education of females as well as more labour-market-oriented education; the greater importance of market forces that dissipate discrimination associated with factors such as global competition, deregulation and privatization; the greater emphasis on “brain over brawn” facilitated by technological change and especially the IT revolution; the dissipation of stereotypes as more women enter the labour market; more-equitable divisions of labour within the household; a growth in demand for female-dominated occupations, such as health and education, and a decline in demand in male-dominated jobs, such as blue-collar manufacturing the decline of male unionization and a relative increase of female unionization; and possibly increased anti-discrimination legislation.

The recent research shows that male-female pay gap is declining over time but at a slower rate. Claudia Goldin (2014) has argued that the gender gap in pay would be reduced further and perhaps eliminated if market forces can foster more flexibility work schedules and better working environment for female workers.

Answers to End-of-Chapter Problems

1. (LO2)
 - a) The job evaluation score is supposed to capture many of the job attributes, such as the level of human capital required, and the various amenities and dis-amenities, such as working conditions, level of experience, etc. Whether you are a man or a woman in a female-dominated occupation, your pay is estimated by the equation for Y_f . This equation generates estimates of actual pay based on averages. It does not predict the exact level of pay for everyone who works in that occupation.

Simply insert the value of 200 into the equation to yield: $4,000 + 80(200) = \$20,000$.

- b) The idea behind pay equity is that anyone, man or woman, who works in a female-dominated occupation should earn a wage based on the pay scale (with its rates per year of experience, educational degree, etc.) of the male-dominated occupation. The pay scale is described by the equation for Y_m . Each point awarded by the job evaluator is worth \$100 in pay rather than \$80 in pay, and the amount of pay associated with 0 points (sometimes called the baseline case) has been raised from \$4,000 to \$5,000. The calculation is: $5,000 + 100(200) = \$25,000$. The magnitude of the pay equity award is \$5,000. Geometrically, this is represented by a vertical movement up to the higher pay line, holding the number of points associated with the job constant.

2. (LO2)

- a) Apply the formulas $Y_m = 5,000 + 100 \cdot P_m = 5,000 + 100(300) = \$35,000$ and $Y_f = 4,000 + 80 \cdot P_f = 4,000 + 80(200) = \$20,000$. The male-dominated jobs have a score of 300 points, and the female-dominated jobs have a score of 200 points. The female-dominated jobs are not remunerated as well, given a constant number of job evaluation points, as reflected in the discrepancy of the coefficients (100 versus 80). Furthermore, the compensation that is associated with 0 points is higher for male-dominated jobs (\$5,000 versus \$4,000).
- b) If women were working in male-dominated jobs, they would earn $5,000 + 100 \cdot P_m = 5,000 + 100(200) = \$25,000$, but females earn only $Y_f = 4,000 + 80 \cdot P_f = 4,000 + 80(200) = \$20,000$; thus \$5,000 of the gap is due to discrimination. \$10,000 of the gap is due to the fact that the female-dominated jobs have lower job evaluation points, and this effect is not considered to be discriminatory.
- c) If men were working in female-dominated jobs, they would earn $4,000 + 80 \cdot P_m = 4,000 + 80(300) = \$28,000$, but they actually earn $Y_m = 5,000 + 100 \cdot P_m = 5,000 + 100(300) = \$35,000$, which leaves a value of \$7,000 for the discrimination component. We thus obtain two fairly close but different estimates for the discrimination component. The first is obtained by placing the women in male-dominated jobs (i.e., inserting the value of the job evaluation points for female-dominated jobs into the equation and remunerating them according to the pay scale for male-dominated jobs), while the second is obtained by the opposite.

3. (LO1)

- a) The profit function is expressed as total revenue minus total cost. For simplicity, in this case we assume that labour is the only factor of production, and that it is totally variable.

$$\Pi = P \cdot Q - W_f(1 + d_f)L_f - W_m(1 - d_m)L_m$$

This function is maximized with respect to L_F , the amount of female labour hired, and L_M , the amount of male labour hired. The labour demand function for men is: $MRP_M = W_M(1 - d_M)$ and for women is: $MRP_F = W_F(1 + d_F)$. Compare these equations to the normal equilibrium condition in which the marginal revenue product is equal to the wage. The impact of the discrimination parameter d_F is to drive a wedge between the MRP of women and the wage rate in equilibrium, which will lower the demand curve for women. The impact of the nepotism parameter d_M is to narrow the distance between the MRP of men and the wage rate in equilibrium, which will raise the demand curve for men.

- b) By equating MRP_M and MRP_F , we can replace them with MRP. Using the equation above and applying transitivity, $MRP = W_M(1 - d_M) = W_F(1 + d_F)$, or $W_M(1 - 0.1) = W_F(1 + 0.1)$, or $W_F / W_M = 0.9/1.1$.
 - c) Employers are willing to hire females provided that their MRP exceeds their wage by 10% or more. Women have to more than pay their way by a factor of 1.1 (instead of a neutral coefficient of 1.0). Employers are willing to hire males provided that their MRP is at least 90% of their wage. Male workers do not have to pay all of their way — only by a factor of 0.9 (instead of 1.0). In this application, discrimination has its limits; this is not a case of discrimination at all costs. Even though the employer prefers male workers, if the productivity of the male workers falls below the 0.9 threshold, they will not be hired. Even though the employer has a grudge against female workers, if their productivity rises above the 1.1 threshold, they will be hired.
4. (LO1) This question is a lot like problem 3, except that no information on the utility functions for these firms is given.
- a) Under normal conditions, firms maximize profits by paying their workers the value of their marginal revenue products. The only argument in the firm's utility function is profits in this case. It is the goal of profit maximization that leads to workers being paid their marginal revenue products. We are told, however, that there is another argument in the firm's utility function, namely the number of women in its workforce. The marginal utility of employing women is negative. Comparing this to the base case in which the firm's utility function is its profit function, the disutility of employing women has to drive a wedge between the marginal revenue product of women and the wages that they are paid, such that wages are lower.
 - b) We are told that the marginal revenue products for men and women are equal. Therefore, by taking the result from part (a) and applying transitivity, we obtain the desired result.